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COURSE: **CHE 312: Transport Phenomena II**

Spring Semester 2020

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8/10

Quiz #2

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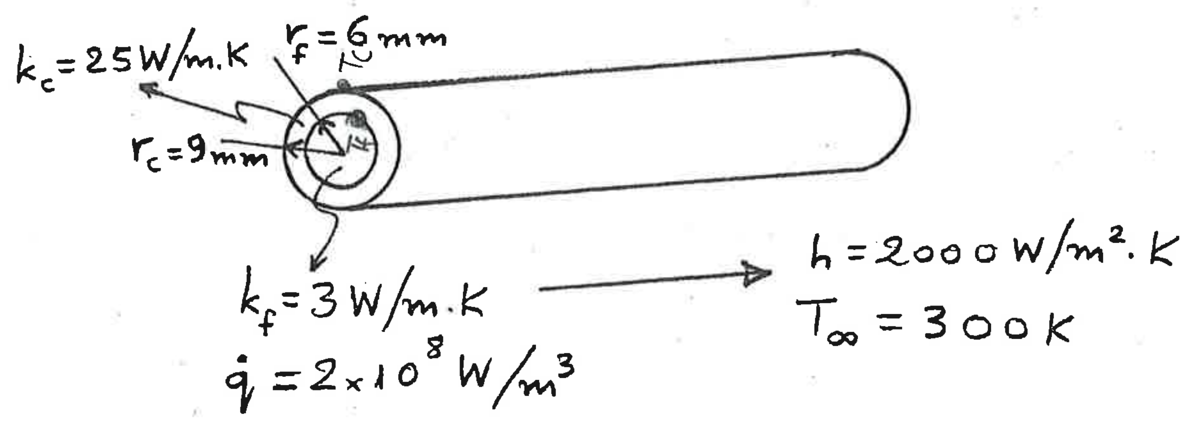
Closed Notes and Book

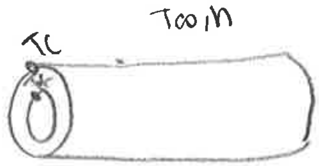
Duration: 25 minutes

A nuclear reactor fuel element consists of a solid cylindrical pin of radius r_f and thermal conductivity k_f. The fuel pin is in good contact with a cladding material of outer radius r_c and thermal conductivity k_c Consider steady-state conditions for which uniform heat generation occurs within the fuel at a volumetric rate $\dot{q}$ and the outer surface of the cladding is exposed to a coolant that is characterized by a temperature T_∞ and a convection coefficient h.

a) Obtain equations for the temperature distributions $T_f(r)$ and $T_c(r)$ in the fuel and cladding, respectively. Express your results exclusively in terms of the foregoing variables.

b) Consider a uranium oxide fuel pin for which k_f = 3 W/m·K and r_f = 6 mm and cladding for which k_c = 25 W/m·K and r_c = 9 mm. If $\dot{q} = 2 \times 10^8$ W/m^3, h = 2000 W/m^2·K, and T_∞ = 300 K, what is the maximum temperature in the fuel element?





Assumptions?

a) Equations for temp dis

Inner Temperature Distribution: $r = 0$ to r_f

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) + \dot{q} = 0$$

$$\int \frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) = \int -\dot{q}r$$

$$kr\frac{\partial T}{\partial r} = \frac{-\dot{q}r^2}{2} + C_1$$

$$\int \frac{\partial T}{\partial r} = \int -\frac{\dot{q}r}{2k} + C_1/kr$$

$$T(r) = -\frac{1}{4}\frac{\dot{q}r^2}{k} + \frac{C_1}{k}\ln|r| + C_2$$

Boundary condition:

we have:

① $T(r_f) =$

also symmetry boundary condition

② $-k\left.\frac{dT}{dr}\right|_{r=0} = 0 \rightarrow$ Thus $\boxed{C_1 = 0}$

Therefore, $T_f(r) = -\frac{1}{4}\frac{\dot{q}r^2}{k} + C_2$

from B.C ①, we know that $T(r_f)$

so:

$$\frac{\dot{q}r_f}{2}\frac{(r_c - r_f)}{k} + \frac{\dot{q}r_f}{2h} + T_\infty = -\frac{1}{4}\frac{\dot{q}r_f^2}{k} + C_2$$

$$C_2 = \frac{\dot{q}r_f}{2}\frac{(r_c - r_f)}{k} + \frac{\dot{q}r_f}{2h} + T_\infty + \frac{1}{4}\frac{\dot{q}r_f^2}{k}$$

Therefore,

a) $$\boxed{T_f(r) = -\frac{1}{4}\frac{\dot{q}r^2}{k} + \frac{\dot{q}r_f}{2}\frac{(r_c - r_f)}{k} + \frac{\dot{q}r_f}{2h} + T_\infty + \frac{1}{4}\frac{\dot{q}r_f^2}{k}}$$

Energy Balance:

$$\frac{\dot{q}(2\pi r_f^2)L}{2\pi r L} = k\frac{(T_f - T_c)}{3 \cdot 10^{-3}\,m} = h(T_c - T_\infty)$$

$$\frac{\dot{q}r_f}{2} = \frac{k(T_f - T_c)}{r_c - r_f} = h(T_c - T_\infty)$$

$$\frac{\dot{q}r_f}{2} = hT_c - hT_\infty$$

$$\frac{\frac{\dot{q}r_f}{2} + hT_\infty}{h} = T_c = \text{outer}$$

Thus:

$$\frac{\dot{q}r_f}{2} = \frac{k\left(T_f - \left(\frac{\dot{q}r_f}{2} + hT_\infty\right)/h\right)}{r_c - r_f}$$

$$\frac{\frac{\dot{q}r_f}{2}(r_c - r_f)}{k} + \left(\frac{\dot{q}r_f}{2} + hT_\infty\right)/h = T_f = \text{inner}$$

Boundary Conditions for T_c

Outer Temp distribution: $T_c(r)$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) = 0$$

$$kr\frac{\partial T}{\partial r} = C_1$$

$$\int \frac{\partial T}{\partial r} = \int C_1/kr$$

$$T_c(r) = \left(\frac{C_1}{k}\right)\ln r + C_2$$

BC ③ $T_c(r_c) =$

$$\frac{\frac{\dot{q}r_f}{2} + hT_\infty}{h} = \frac{C_1}{k}\ln r_c + C_2$$

$$\frac{\dot{q}r_f}{2h} + T_\infty = -\frac{\dot{q}r_f(r_c - r_f)}{k\,2\ln(r_f)}\ln r_c + C_2$$

$$C_2 = \frac{\dot{q}r_f}{2h} + T_\infty + \frac{\dot{q}r_f(r_c - r_f)\ln(r_c)}{2k\ln(r_f)}$$

④ $T_c(r_f) =$

$$\frac{\left(\frac{\dot{q}r_f}{2}\right)(r_c - r_f)}{k} + \frac{\dot{q}r_f}{2h} + T_\infty = \frac{C_1}{k}\ln(r_f) + C_2$$

$T_c(r_c) - T_c(r_f) =$

$$\frac{\dot{q}r_f}{2h} + T_\infty - \left(\frac{\dot{q}r_f}{2}\right)\frac{(r_c - r_f)}{k} - \cancel{\frac{\dot{q}r_f}{2h}} - \cancel{T_\infty} - \frac{C_1}{k}\ln(r_f) + \frac{C_1}{k}$$

$$\frac{-\frac{\cancel{k}\dot{q}r_f}{2}\frac{(r_c - r_f)}{\cancel{k}}}{\ln(r_c)} = C_1 = -\frac{\dot{q}r_f(r_c - r_f)}{2\ln(r_f)}$$

BC:

we know T_c and T_f

Using calculated C_1 and C_2 for T_c:

Thus,

a) $$\boxed{T_c(r) = \frac{-\dot{q}r_f(r_c - r_f)}{k \cdot 2\ln(r_f)}\ln|r| + \frac{\dot{q}r_f}{2h} + \frac{\dot{q}r_f(r_c - r_f)\ln(r_c)}{2k\ln(r_f)} + T_\infty}$$

Part b: max temp would be at $r = 0$, for temp distribution

For $T_f(r)$:

$$T_f(0) = -\frac{1}{4}(0) + \frac{(2 \cdot 10^8\,W/m^3)(6 \cdot 10^{-3}\,m)}{2}\frac{(3 \cdot 10^{-3}\,m)}{(3\,W/mK)} + \frac{(2 \cdot 10^8)(6 \cdot 10^{-3}\,m)}{2(2000)} + 300 + \frac{1}{4}\frac{(2 \cdot 10^8)(6 \cdot 10^{-3})^2}{3000} = \boxed{1200.9\,K} \leftarrow \text{max temp}$$

express 2 sig. figs